

HW3 – Knowledge Representation and Reasoning (Propositional Logic, CNF, First-Order Logic)

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1) Propositional Reasoning in a Tiny Wumpus World (30 pts)

(a) Truth Table (10 pts)

The knowledge base consists of:

1. B_{11} (there is a breeze in (1,1))
2. $B_{11} \leftrightarrow (P_{12} \vee P_{21})$

Since B_{11} must be true and the biconditional must hold, KB is true only when $B_{11} = T$ and at least one of P_{12} or P_{21} is true. Note that P_{11} and P_{22} do not appear in KB, so they don't affect its truth value.

B_{11}	P_{11}	P_{12}	P_{21}	P_{22}	KB
F	F	F	F	F	F
F	F	F	T	F	F
F	F	T	F	F	F
F	F	T	T	F	F
F	T	F	F	F	F
F	T	F	T	F	F
F	T	T	F	F	F
F	T	T	T	F	F
T	F	F	F	F	F
T	F	F	T	F	T
T	F	T	F	F	T
T	F	T	T	F	T
T	T	F	F	F	F
T	T	F	T	F	T
T	T	T	F	F	T
T	T	T	T	F	T

Note: P_{22} is fixed to F in this table since it does not appear in KB. Changing P_{22} to T would not affect any KB values.

(b) Does KB entail P_{12} ? (10 pts)

Answer: $\boxed{\text{KB} \not\models P_{12}}$

There are rows where KB is true but P_{12} is false (for example, row 10 where $P_{21} = T$ makes KB true while $P_{12} = F$). So KB does not entail P_{12} .

(c) Does KB entail $\neg P_{12}$? (10 pts)

Answer: $\boxed{\text{KB} \not\models \neg P_{12}}$

There are rows where KB is true but P_{12} is also true (rows 11, 12, 15, 16). In these cases $\neg P_{12}$ is false, so KB does not entail $\neg P_{12}$.

2) Conjunctive Normal Form (CNF) in the Wumpus World (30 pts)

(a) Propositional sentence (5 pts)

“If there is no breeze in (1, 1), then there is no pit in (1, 2) and no pit in (2, 1).”

$$\boxed{\neg B_{11} \rightarrow (\neg P_{12} \wedge \neg P_{21})}$$

(b) CNF conversion (25 pts)

Starting formula:

$$\neg B_{11} \rightarrow (\neg P_{12} \wedge \neg P_{21})$$

Step 1: Eliminate implication ($\alpha \rightarrow \beta \equiv \neg\alpha \vee \beta$)

$$\neg(\neg B_{11}) \vee (\neg P_{12} \wedge \neg P_{21})$$

Step 2: Double negation elimination ($\neg\neg\alpha \equiv \alpha$)

$$B_{11} \vee (\neg P_{12} \wedge \neg P_{21})$$

Step 3: Distribute $\vee$ over $\wedge$ ($\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$)

$$(B_{11} \vee \neg P_{12}) \wedge (B_{11} \vee \neg P_{21})$$

Final CNF:

$$\boxed{(B_{11} \vee \neg P_{12}) \wedge (B_{11} \vee \neg P_{21})}$$

3) First-Order Logic (FOL) Representation (40 pts)

(a) 10 pts

“Every CS student takes CS404 or CS408.”

$$\boxed{\forall x (Enrolled(x, CS) \rightarrow (Takes(x, CS404) \vee Takes(x, CS408)))}$$

(b) 10 pts

“No one likes greedy people.”

$$\boxed{\forall x \forall y ((Person(x) \wedge Greedy(y)) \rightarrow \neg Likes(x, y))}$$

(c) 10 pts

“If a hard-working student will pass a course s/he is taking.”

(Interpreting this as: a hard-working student passes any course they take.)

$$\boxed{\forall x \forall y ((Hardworking(x) \wedge Takes(x, y)) \rightarrow Pass(x, y))}$$

(d) 10 pts

“Half siblings have a different mom or dad.”

Half-siblings share exactly one parent, meaning either their mothers differ or their fathers differ.

$$\begin{aligned} \forall x \forall y \Big(& Halfsibling(x, y) \rightarrow \neg \exists m (Mother(m, x) \wedge Mother(m, y)) \\ & \vee \neg \exists f (Father(f, x) \wedge Father(f, y)) \Big) \end{aligned}$$

Alternatively:

$$\begin{aligned} \forall x \forall y \forall m_1 \forall m_2 \forall f_1 \forall f_2 \Big(& (Halfsibling(x, y) \wedge Mother(m_1, x) \wedge Mother(m_2, y) \\ & \wedge Father(f_1, x) \wedge Father(f_2, y)) \\ & \rightarrow (m_1 \neq m_2 \vee f_1 \neq f_2) \Big) \end{aligned}$$